

Kawsar Ahemmed Bappy

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Dhaka, Bangladesh

Work Experience

Technonext Software Limited

Dhaka, Bangladesh

Software Engineer (Mid) (July 2025–Present)

- Delivered geospatial APIs (geocode, reverse-geocode, radius search, autocomplete) for MapNest. Integrated Elasticsearch with OSM/OSRM/Nominatim. Achieved **sub-100ms p99 latency** for 1.3M+ records.
- Optimized Elasticsearch for Autocomplete API through field filtering and usage analysis. Achieved **87% faster queries, 97.5% less bandwidth, 30% smaller indexes, and 20% faster indexing**.
- Participated in designing distributed microservices (Go, gRPC, RabbitMQ) for order processing. Optimized async workflows to **reduce latency by 32%** (2.8s → 1.9s) under 10,000+ concurrent orders.
- Architected and built centralized User Management & RBAC microservices in Go using gRPC, OIDC/OAuth2 social login, JWT/token reuse detection, and Redis bitmap-based authorization; migrated auth from monoliths, improving security and reducing permission-check **latency by 60% across 5+ platforms**.
- Created Geo-Syncer, an async data pipeline with PostgreSQL triggers and RabbitMQ. Automated Nominatim-to-Elasticsearch syncs. **Eliminated 1-hour manual workflows** with 99.9% data consistency.
- Initiated and led internal knowledge-sharing sessions on Redis, Go design, clean architecture, and PostgreSQL internals; mentored new engineers on SOLID principles, codebase practices, and production debugging

Vivasoft Limited

Dhaka, Bangladesh

Software Engineer (L-1) (March 2024–July 2025)

- Built AI-powered job crawler for WeSelect using Python/FastAPI and Go/Gin. Deployed hybrid GPT-3.5/GPT-4o architecture. Processed **1,000+ daily listings** with **40% cost savings** vs. third-party tools.
- Refactored Pi-Inventory legacy codebase with clean architecture and SOLID principles. **Reduced code by 30-40%**. Improved maintainability and cut onboarding time.
- Boosted database performance via Redis caching, N+1 query fixes, and GORM optimization. **Improved response times by 20-30%** across multiple services.
- Containerized microservices with Docker and streamlined CI/CD pipelines. **Cut deployment time from 45 minutes to 8 minutes**.
- Led Financfy mobile app migration from Flutter 3.10.0 to 3.24.5. Fixed 50+ breaking changes. Implemented type-safe RBAC using BLoC. Expanded test coverage.

Vivasoft Limited

Dhaka, Bangladesh

Associate Software Engineer (February 2023–March 2024)

- Built RESTful APIs for inventory management system using Go and Gin framework. Resolved 30+ production bugs in inventory module, improving system stability.
- Contributed to open-source Golang Bootcamp in Bangla with beginner-friendly docs and examples. Simplified learning curve for complex Go concepts.

Technical Skills

Languages: Go, Dart, Python

Frameworks: Gin, Echo, Flutter

Databases: PostgreSQL, MySQL, Redis, GORM

Search & Messaging: Elasticsearch, RabbitMQ

Auth & Security: JWT, OIDC, OAuth 2.0, RBAC

Tools & Platforms: Docker, AWS (EC2, S3), Kibana, Grafana, Elastic APM, Prometheus

Concepts: Clean Architecture, SOLID, Microservices, Event-Driven Systems, CDC, Saga Pattern, Redis Bitmaps, RBAC

Projects

MapNest Geospatial Platform (*Go, Elasticsearch, OSM, OSRM, Nominatim, gRPC*)

Production mapping platform serving geocoding and location search services for internal applications across the organization.

- Architected Elasticsearch spatial indexing strategy using `geo_point` and `geo_shape` data types, implementing custom scoring functions for relevance-based location ranking.

- Developed radius search algorithm combining Haversine distance calculation with Elasticsearch's geo_distance queries for accurate proximity-based results.
- Built fallback architecture integrating multiple data providers (OSM, Google Places) with circuit breaker pattern to ensure 99.9% API availability.
- Built some ETL pipelines to seed and sync PostgreSQL data into Elasticsearch for MapNest, enabling full geospatial search indexing across 1.3M+ records

Foodi Mart (*Go, gRPC, RabbitMQ, PostgreSQL, Docker*)

Distributed microservices ecosystem powering inventory management and order fulfillment for high-traffic e-commerce operations.

- Implemented saga pattern for distributed transactions across Order, Inventory, and Payment services, ensuring eventual consistency with compensating actions.
- Designed RabbitMQ topic exchange architecture with dead-letter queues for fault-tolerant message processing during traffic spikes.
- Built inventory reservation system with optimistic locking and TTL-based cleanup to prevent overselling during concurrent checkout operations.

Geo-Syncer (*Go, PostgreSQL, RabbitMQ, Elasticsearch, CDC*)

Event-driven data synchronization pipeline maintaining consistency between PostgreSQL (source of truth) and Elasticsearch (search index).

- Designed PostgreSQL trigger-based CDC (Change Data Capture) system writing to changelog tables, avoiding expensive polling mechanisms.
- Implemented batch processing with configurable window sizes, balancing real-time updates with Elasticsearch bulk indexing efficiency.
- Built idempotent sync handlers with versioning strategy to handle out-of-order messages and ensure eventual consistency across distributed systems.

WeSelect AI Job Crawler (*Python, FastAPI, Go, Gin, PostgreSQL, OpenAI*)

AI-powered career intelligence platform that automatically extracts and structures job postings from multiple sources for market analysis.

- Designed hybrid LLM strategy balancing GPT-3.5 for bulk extraction and GPT-4o for complex parsing, implementing retry logic and fallback mechanisms to ensure data quality.
- Built concurrent crawler engine in Go with rate limiting, proxy rotation, and anti-bot detection handling for reliable data collection at scale.
- Implemented structured prompt engineering techniques to extract 15+ job attributes with 92% accuracy.

Financfy SaaS Suite (*Go, Gin, PostgreSQL, MySQL, Redis, Flutter, BLoC, Firebase*)

Multi-tenant financial management platform serving SMEs with inventory tracking, asset management, and mobile operations.

- Contributed to multiple backend modules following clean architecture principles, enabling independent testing and loose coupling.
- Implemented hierarchical RBAC system with permission inheritance and resource-level access control, supporting complex organizational structures.
- Designed Flutter state management using BLoC pattern with event-driven architecture, separating business logic from UI for enhanced testability.
- Built custom Flutter route guards based on permissions and persistent navigation state across app restarts.

Problem Solving & Activities

- Solved 400+ problems on LeetCode and other online judges
- Advanced Learning Algorithms (Coursera)
- SQL (Intermediate) (HackerRank)
- Docker Essentials (KodeKloud)

Education

B.Sc. in Software Engineering

Shahjalal University of Science and Technology | February 2019 – February 2024 | Sylhet, Bangladesh